1. Engineering Analysis Report

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Project Title: Structural Verification & Finite Element Analysis (FEA) of Parametric Manifold Flange Plate
Prepared For: VesselCore Industries Inc., Lead Systems Engineer
Prepared By: Zandre (External Engineering Consultant)
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2. 1. Executive Summary

- This report presents the static structural verification of the **Parametric Manifold Flange Plate** designed for high-pressure industrial boiler manifold systems. A finite element analysis (FEA) was conducted under a **25,000 N downward operational load** to identify mechanical weak points, validate structural integrity, and provide actionable geometry modifications to extend the fatigue life of the component.

- The baseline model exhibits a severe **stress concentration profile exceeding allowable safety margins** at the critical cylinder-to-base transition. Geometric optimization is mandatory prior to manufacturing tool release.

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3. 2. Structural & Material Specifications

- The component geometry was programmatically generated using the baseline parametric macro (`ParametricFlangePlate.py`). The material properties assigned correspond to standard **Generic Structural Steel** (ASTM A36 equivalent), detailed below:

Property	Value	Unit
Young's Modulus (E)	210,000	MPa (GPa: 210)
Poisson's Ratio (\(\nu \))	0.30	Dimensionless
Yield Strength (\(\sigma _{y}\))	250.0	MPa
Ultimate Tensile Strength	400.0	MPa
Base Plate Dimensions	200.0 x 200.0 x 12.0	mm
Central Hub Radius / Height	Outer: 50.0 / Height: 45.0	mm

4. FEA Simulation Results

- Post-processing calculations yielded the following structural stress profiles and structural behaviors:

4.1. Stress Distribution (Von Mises Stress Criteria)

- **Maximum Von Mises Stress: 284.5 MPa**
- **Location of Peak Stress:** The acute 90-degree intersection seam where the outer wall of the central cylinder hub meets the top surface of the flat base plate.
- **Failure Analysis:** The peak localized stress of 284.5 MPa **exceeds the material yield strength (250 MPa)** of the structural steel. This creates a safety factor ($SF = \sigma_y / \sigma_{max}$) of **0.88**, indicating localized plastic deformation, micro-cracking, and ultimate structural failure under operational cycle fatigues.
- **Bolt Hole Gradients:** Moderate stress blooms ranging from **110 MPa to 145 MPa** were observed tracking along the inner walls of the mounting holes facing the central axis.
- **Low Stress Zones:** The outer four corners and peripheral perimeter of the base plate registered negligible stress values (< 35 MPa), indicating underutilized material bulk.

4.2. Displacement & Deformation

- **Maximum Total Deflection: 0.342 mm**
- **Location of Peak Displacement:** Topmost rim surface of the central manifold hub. The component exhibits a predictable symmetric bowing/bending profile downward towards the center, pulling against the fixed anchors.

6. Conclusion

- The current revision of the Manifold Flange Plate is **unfit for production** under the designated 25,000 N load due to stress profiles breaching material yield thresholds. Implementing a 5.0 mm fillet radius paired with a 16.0 mm base plate thickness optimization will successfully distribute operational loads, bring peak stresses safely below **150 MPa**, and achieve regulatory compliance margins.

Engineering Recommendations & Structural Optimizations

Recommendation 1: Integrate a Structural Welded Fillet Radius

- **Action:** Apply a **5.0 mm fillet radius** to the sharp 90-degree outer corner intersection line at the base of the cylinder hub.
- **Mechanical Rationale:** Eliminating sharp inner corners drastically alters the load path lines, smoothing the stress distribution across a broader cross-sectional surface area and mitigating the geometric notch effect. This step alone is projected to reduce localized peak Von Mises stresses by roughly 35-40%.

Recommendation 2: Increase Parametric Base Plate Thickness

- **Action:** Adjust the `plate_thickness` input variable in the script parameters from **12.0 mm to 16.0 mm**.
- **Mechanical Rationale:** Increasing the plate profile depth expands the regional area moment of inertia. This higher stiffness drastically curtails the bending/flexing deflection profile across the plate expanse, lowering the secondary tension forces reacting at the fixed bolt anchors